



Brief paper

Delay-independent stabilization of a class of time-delay systems via periodically intermittent control[☆]



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ABSTRACT

The problem of delay-independently periodically intermittent stabilization for a class of time-delay systems is examined. First, the stability of the considered periodically intermittently controlled time-delay systems is analyzed by using the piecewise switching-time-dependent Lyapunov function/functional. The introduced Lyapunov function/functional is nonincreasing at switching instants, which can guarantee the exponential stability of the considered systems irrespective of the sizes of the state delays. Next, based on the newly established stability criteria, sufficient conditions for the existence of delay-independently periodically intermittent state-feedback controllers are derived. Finally, two illustrative examples are presented to show the validity of the obtained results.

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1. Introduction

The last decades have witnessed an increasing interest in intermittent control due to its potential applications in various fields such as civil/structural engineering, chemical engineering, psychology, and communication. Intermittent control is a discontinuous feedback control, which fills the gap between the two extremes of continuous and impulsive control in that the control signal is implemented intermittently. Compared with continuous control, intermittent control can efficiently save energy and give more flexibility to the designer. On the other hand, compared with impulsive control, intermittent control has the advantage of easy implementation in engineering practice because it does not have to change the state instantaneously. Therefore, the intermittent control scheme integrates the merits of continuous and impulsive control. According to the way that the controller is activated, the intermittent control can be categorized into event-driven

intermittent control and time-driven intermittent control. For the former, the control action is activated only when a pre-given event occurs (Gawthrop, Neild, & Wagg, 2012; Gawthrop & Wang, 2009a,b; Zochowski, 2000), while for the latter, the controller is activated at a sequence of fixed finite time intervals. Several types of time-driven intermittent control have been considered in the literature. For example, there are intermittent predictive control (Ronco, Arsan, & Gawthrop, 1999), delayed feedback control with controller failure (Sun, Liu, Rees, & Wang, 2008), intermittent feedback for model-based networked control systems (Estrada & Antsaklis, 2010; Garcia & Antsaklis, 2013), sampled-data control with control packet loss (Chen & Zheng, 2012; Zhang & Yu, 2010), intermittent impulsive control (Liu, Shen, & Zhang, 2011), distributed consensus of multi-agent systems with intermittent communications (Wen, Duan, Ren, & Chen, 2014), and periodically intermittent feedback control (Chen, Zhong, Jiang, & Lu, 2014; Hu, Yu, Jiang, & Teng, 2010; Huang, Li, & Han, 2009; Huang, Li, & Liu, 2008; Huang, Li, Yu, & Chen, 2009; Li, Liao, & Huang, 2007; Song & Huang, 2015; Wang, Hao, & Zuo, 2010; Xing, Jiang, & Hu, 2013; Yang & Cao, 2009; Yu, Hu, Jiang, & Teng, 2011).

In the framework of periodically intermittent feedback, two distinct modes of operation are involved: the closed-loop mode for the time intervals $[k\omega, k\omega + \delta)$, and the open-loop mode for the time intervals $[k\omega + \delta, (k+1)\omega)$, where the parameters ω and δ denote the control period and the control width, respectively. So the periodically intermittently controlled system can be viewed as a switched system in which the closed-loop mode and the open-loop mode operate in alternating manner. Correspondingly, the

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stability analysis of a periodically intermittently controlled system boils down to the stability analysis of a time-driven switching system consisting a stable subsystem and an unstable subsystem. In the previous results (Hu et al., 2010; Huang, Li, & Han, 2009; Huang et al., 2008; Huang, Li, Yu et al., 2009; Li et al., 2007; Wang et al., 2010; Xing et al., 2013; Yang & Cao, 2009; Yu et al., 2011), the stability of periodically intermittently controlled delayed neural networks was studied. The stability analysis was carried out by applying common Lyapunov function/functional based techniques in which two distinct modes share the same Lyapunov function/functional. The basic idea is to obtain the decay rate estimate of the closed-loop mode and the increase rate estimate of the open-loop mode with the aid of delay differential inequalities. Then by suppressing the increase rate of the open-loop mode using the decay rate of the closed-loop mode, the stability criteria are established. It is well-known that the existence of a common Lyapunov function is too rigorous to stability analysis of many switched systems (Chen & Zheng, 2010; Sun, Zhao, & Hill, 2006; Zhao & Hill, 2008). So these results may be conservative. Recently, Chen et al. (2014) proposed the piecewise Lyapunov function/functional based approaches for stability analysis and synthesis of periodically intermittently controlled time-delay systems. These approaches can make good use of mode information and reduce the conservatism inherent in the common Lyapunov function/functional based methods. However, like the results given in Huang et al. (2008), Huang, Li, and Han (2009); Huang, Li, Yu et al. (2009), Li et al. (2007), Wang et al. (2010) and Yang and Cao (2009), the stability criteria derived in Chen et al. (2014) also require that the control width should be strictly larger than the size of the state delay. Although the theoretical results presented in Hu et al. (2010), Xing et al. (2013) and Yu et al. (2011) removed this constraint, their stability conditions still depend upon the delay bounds. This means that all the existing stability results for periodically intermittently controlled time-delay systems are only suitable to the case where the delay bounds are known even if the original time-delay systems can be delay-independently stabilizable. These observations motivate the study of the following question: for a delay-independently stabilizable time-delay system, what conditions can guarantee the existence of delay-independently periodically intermittently state-feedback controllers? It should be pointed out that this question cannot be solved by the existing exponent estimation based methods. This is because the exponential decay/increase rate of a time-delay system usually depends on the delay bounds.

The purpose of this paper is to study the problem of delay-independent stabilization of a class of time-delay systems via periodically intermittently control. Piecewise switching-time-dependent Lyapunov functions/functionals are introduced to analyze the stability of the periodically intermittently controlled time-delay systems. An important feature of the introduced Lyapunov functions/functionals is the nonincreasing property at the switching instants. Such feature makes the stability analysis of the periodically intermittently controlled time-delay system like the stability analysis for a time-delay system without switching. As a result, delay-independent stability criteria are obtained for the periodically intermittently controlled time-delay systems. Then using the linear matrix inequalities (LMIs) techniques, sufficient conditions for the existence of delay-independently periodically intermittent state-feedback controllers are established.

The rest of the paper is organized as follows. In Section 2, some basic definitions and preliminaries are presented. In Section 3, stability criteria for the considered systems are developed by using the piecewise switching-time-dependent Lyapunov function/functional based methods. Section 4 provides sufficient conditions on the existence of delay-independently periodically intermittent state-feedback controllers. In Section 5, two numerical examples are given to demonstrate the efficiency of the proposed methods. Finally, some concluding remarks are made in Section 6.

2. System description and preliminaries

In the sequel, if not explicitly, matrices are assumed to have compatible dimensions. The notation $M > (\geq, <, \leq) 0$ is used to denote a symmetric positive-definite (positive-semidefinite, negative, negative-semidefinite) matrix. I stands for an identity matrix of suitable dimension. $\|\cdot\|$ refers to the Euclidean vector norm. $\mathbb{N}$ represents the set of nonnegative integers, that is, $\mathbb{N} = \{0, 1, 2, \dots\}$. For $\tau > 0$, let $C([-\tau, 0], \mathbb{R}^n)$ denote the space of bounded, continuous functions $x : [-\tau, 0] \mapsto \mathbb{R}^n$ with norm $\|x\|_\tau = \max_{-\tau \leq \theta \leq 0} \|x(t + \theta)\|$. If $y \in C([-\tau, \alpha], \mathbb{R}^n)$ with $\alpha > 0$ and $t \in [0, \alpha]$, then $y_t \in C([-\tau, 0], \mathbb{R}^n)$ is defined by $y_t(\theta) = y(t + \theta)$, $\theta \in [-\tau, 0]$.

Consider a class of time-delay systems of the form

$$\begin{aligned} \dot{x}(t) &= \sum_{l=0}^1 [C_l x(t - \tau_{lt}) + A_l f(x(t - \tau_{lt}))] \\ &\quad + Bu(t), \quad t > 0, \\ x(t) &= \phi(t), \quad -\bar{\tau} \leq t \leq 0, \end{aligned} \quad (1)$$

where $x(t) \in \mathbb{R}^n$ and $u(t) \in \mathbb{R}^m$ denote the vectors of the system state and the input at time t ; $C_l, A_l \in \mathbb{R}^{n \times n}$, $l = 0, 1$, are constant matrices; $f(x) = (f_1(x_1), f_2(x_2), \dots, f_n(x_n))^T$ is a continuous map from $\mathbb{R}^n$ to $\mathbb{R}^n$; $\tau_{0t} = 0$, and $\tau_{1t} = \tau(t)$, where $\tau(t)$ is a time-varying function satisfying $0 \leq \tau(t) \leq \bar{\tau}$; $\phi \in C([-\bar{\tau}, 0]; \mathbb{R}^n)$ is the initial function. Throughout this paper, assume that the continuous map $f(\cdot)$ satisfies the following condition.

Assumption (A1) There exist scalars κ_i^- and κ_i^+ such that for any $s_1, s_2 \in \mathbb{R}$, $s_1 \neq s_2$,

$$\kappa_i^- \leq \frac{f_i(s_1) - f_i(s_2)}{s_1 - s_2} \leq \kappa_i^+, \quad i = 1, 2, \dots, n.$$

Moreover, we consider the following two cases for the time-varying delay $\tau(t)$.

Case 1: $\tau(t)$ is a bounded continuous function satisfying

$$0 \leq \tau(t) \leq \bar{\tau}, \quad \forall t \geq 0,$$

where $\bar{\tau}$ is a scalar representing the upper bound of the time delay.

Case 2: $\tau(t)$ is a bounded differentiable function satisfying

$$0 \leq \tau(t) \leq \bar{\tau}, \quad \dot{\tau}(t) \leq r < 1, \quad \forall t \geq 0,$$

where r is a nonnegative scalar.

For system (1), the following assumption is made throughout the paper.

Assumption (A2): System (1) with $u = 0$ is unstable.

The periodically intermittent linear state-feedback control law takes the following form

$$u(t) = K(t)x(t) \quad (2)$$

with

$$K(t) = \begin{cases} K, & t \in \Delta_{1k} \triangleq [k\omega, k\omega + \delta), \\ 0, & t \in \Delta_{2k} \triangleq [k\omega + \delta, (k+1)\omega), \end{cases}$$

where $K \in \mathbb{R}^{m \times n}$ is the control gain matrix to be designed.

Remark 1. The structure of the intermittent controller (2) implies that $u(t) = 0$ for all $t \in \Delta_{2k}$, $k \in \mathbb{N}$. This assumption makes sense as it provides a good representation of certain events. For example, it can model the controller failure or the purposeful suspension of controllers for equipment maintenance or a decreased bandwidth usage.

With the control law (2), system (1) can be written as

$$\begin{cases} \dot{x}(t) = \sum_{l=0}^1 [\bar{C}_{1l}x(t - \tau_{lt}) + A_l f(x(t - \tau_{lt}))], & t \in \Delta_{1k}, \\ \dot{x}(t) = \sum_{l=0}^1 [\bar{C}_{2l}x(t - \tau_{lt}) + A_l f(x(t - \tau_{lt}))], & t \in \Delta_{2k}, \\ x(t) = \phi(t), & -\bar{\tau} \leq t \leq 0, \end{cases} \quad (3)$$

in which $\bar{C}_{10} = C_0 + BK$, $\bar{C}_{20} = C_0$, and $\bar{C}_{11} = \bar{C}_{21} = C_1$. It is obvious that system (3) is a switched one with two distinct modes. For $t \in \Delta_{1k}$, the system is operating in closed-loop mode. For $t \in \Delta_{2k}$, the system is operating in open-loop mode. In this paper, given (ω, δ) , we will deal with the design of periodically intermittent state-feedback controller (2) such that the resulting closed-loop system (3) is exponentially stable for any $\bar{\tau} \geq 0$.

The following definition and lemmas will be used in deriving the main results of this paper.

Definition 1. The zero solution of (3) is said to be globally exponentially stable (GES) if there exist two positive scalars M and γ such that

$$\|x(t, 0, \phi)\| \leq M \|\phi\|_T \exp\{-\gamma t\}, \quad t \geq 0.$$

Lemma 1 (Chen et al., 2014). For matrices $\mathcal{D} \in \mathbb{R}^{n \times n}$, $\mathcal{E} \in \mathbb{R}^{N \times n}$, $\Gamma_i \in \mathbb{R}^{N \times N}$, $\Gamma_i = \Gamma_i^T$, $X_0, X_i, H_i \in \mathbb{R}^{n \times n}$, $i = 1, 2, \dots, p$, if they satisfy the following inequalities:

$$\begin{bmatrix} \Gamma_i + \mathcal{E}X_0\mathcal{D} + (\mathcal{E}X_0\mathcal{D})^T & (X_i - X_0)\mathcal{D}^T + \mathcal{E}H_i \\ * & -H_i - H_i^T \end{bmatrix} < 0, \quad i = 1, \dots, p,$$

then it holds that

$$\Gamma_i + \mathcal{E}X_i\mathcal{D} + (\mathcal{E}X_i\mathcal{D})^T < 0, \quad i = 1, \dots, p.$$

Lemma 2 (Chen & Zheng, 2007). For any $n \times n$ matrices $X > 0$, U , scalar $\varepsilon > 0$, the following matrix inequality holds:

$$UX^{-1}U^T \geq \varepsilon(U + U^T) - \varepsilon^2 X.$$

The following notation will be used in the subsequent sections:

$$\mathcal{A}_i = [\bar{C}_{i0} \quad \bar{C}_{i1} \quad A_0 \quad A_1], \quad i = 1, 2,$$

$$\delta_1 = \delta, \quad \delta_2 = \omega - \delta,$$

$$L_1 = \text{diag}\{-\kappa_1^+ \kappa_1^-, -\kappa_2^+ \kappa_2^-, \dots, -\kappa_n^+ \kappa_n^-\},$$

$$L_2 = \text{diag}\left\{\frac{\kappa_1^+ + \kappa_1^-}{2}, \frac{\kappa_2^+ + \kappa_2^-}{2}, \dots, \frac{\kappa_n^+ + \kappa_n^-}{2}\right\},$$

$$\eta(t) = \text{col}\{x(t), x(t - \tau(t)), f(x(t)), f(x(t - \tau(t)))\}.$$

3. Stability analysis

This section addresses the delay-independent exponential stability problem of system (3). For this purpose, we introduce several auxiliary functions.

For $t \geq -\bar{\tau}$, define

$$\rho_1(t) = \begin{cases} \frac{t - k\omega}{\delta}, & t \in \Delta_{1k}, k \in \mathbb{N} \\ \frac{t - k\omega - \delta}{\omega - \delta}, & t \in \Delta_{2k}, k \in \mathbb{N} \\ 1, & t \in [-\bar{\tau}, 0), \end{cases}$$

$$\rho_2(t) = 1 - \rho_1(t),$$

$$\sigma(t) = \begin{cases} 1, & t \in [-\bar{\tau}, 0] \cup \left(\bigcup_{k \in \mathbb{N}} \Delta_{1k}\right), \\ 2, & t \in \bigcup_{k \in \mathbb{N}} \Delta_{2k}. \end{cases}$$

Let $\tilde{\rho}_h(t) = \rho_h(t - \tau(t))$ for $t \geq 0$, $h = 1, 2$. For $i \in \{1, 2\}$, and $k \in \mathbb{N}$, define $t_{ik} = \begin{cases} k\omega, & i = 1 \\ k\omega + \delta, & i = 2 \end{cases}$. It is easy to see that for all $i \in \{1, 2\}$ and all $k \in \mathbb{N}$,

$$\Delta_{ik} = [t_{ik}, t_{3-i, k+1-1}), \quad \rho_1(t_{ik}) = 0, \quad \rho_1(t_{ik}^-) = 1. \quad (4)$$

Theorem 1. Given $m \times n$ matrix K , the control period ω , and the control width δ , consider system (3) satisfying assumptions (A1)–(A2). Under Case 1, if for some prescribed positive scalars $\alpha_i, \mu_i, i = 1, 2$, there exist $n \times n$ matrices $P_{ij} > 0, i = 1, 2, n \times n$ positive diagonal matrices $D_{ijhl}, i, j, h = 1, 2, l = 0, 1$, such that the following LMIs hold:

$$\begin{bmatrix} \Omega_{1ijh} & P_{ij}C_1 & \Gamma_{ijh} & P_{ij}A_1 \\ * & \Phi_{ijhp} & 0 & L_2D_{ijh1} \\ * & * & -D_{ijh0} & 0 \\ * & * & * & -D_{ijh1} \end{bmatrix} < 0, \quad i, j, h, p = 1, 2, \quad (5)$$

$$P_{i2} \leq \mu_{3-i} P_{3-i, 1}, \quad i = 1, 2, \quad (6)$$

where $\underline{\mu}_i = \min\{1, \mu_i\}, \bar{\mu}_i = \max\{1, \mu_i\}$, and

$$\begin{aligned} \Omega_{1ijh} &= \left(\frac{\ln \mu_i}{\delta_i} + \alpha_i \bar{\mu}_i\right) P_{ij} + \frac{1}{\delta_i} (P_{i1} - P_{i2}) + P_{ij} \bar{C}_{i0} \\ &\quad + \bar{C}_{i0}^T P_{ij} + L_1 D_{ijh0}, \\ \Phi_{ijhp} &= -\alpha_i \underline{\mu}_p P_{ph} + L_1 D_{ijh1}, \\ \Gamma_{ijh} &= P_{ij} A_0 + L_2 D_{ijh0}, \end{aligned}$$

then system (3) is GES.

Proof. From (5), there exists a positive scalar γ such that LMIs (5) in which $\bar{\mu}_i$ is replaced by $\gamma + \bar{\mu}_i e^{\gamma \bar{\tau}}$ still hold. For illustration purposes, the notation Ξ_{ijhp} is used to denote the matrix on the left side of (5) in which $\bar{\mu}_i$ is replaced by $\gamma + \bar{\mu}_i e^{\gamma \bar{\tau}}$. Then,

$$\Xi_{ijhp} < 0, \quad i, j, h, p = 1, 2. \quad (7)$$

For notational brevity, for $i \in \{1, 2\}$, and $l = \{0, 1\}$, define $D_{il}(t) = \sum_{j,h=1}^2 \rho_j(t) \tilde{\rho}_h(t) D_{ijhl}$, and denote $D_{il}(t)$ by

$$D_{il}(t) = \text{diag}\{d_{il1}(t), d_{il2}(t), \dots, d_{iln}(t)\}.$$

We introduce the following switching-time-dependent Lyapunov function for system (3):

$$V_1(t, x) = \varphi(t) x^T \sum_{j=1}^2 \rho_j(t) P_{\sigma(t)j} x, \quad (8)$$

where $\varphi(t) = \mu_{\sigma(t)}^{\rho_1(t)}$. For any given initial function $\phi \in C([-\bar{\tau}, 0], \mathbb{R}^n)$, set $x(t) = x(t, 0, \phi)$. Let $\epsilon_0 = \|\phi\|_{\bar{\tau}}, \lambda_0 = \min\{\lambda_{\min}(P_{ij}) : i, j = 1, 2\}$, and $\lambda_1 = \max\{\lambda_{\max}(P_{ij}) : i, j = 1, 2\}$. Without loss of generality, we assume that $\epsilon_0 > 0$. Select a positive scalar ϵ_1 such that $\lambda_1 \mu_1 < \lambda_0 \epsilon_1$. Set $W(t) = e^{\gamma t} V_1(t, x(t))$. In the sequel, we will show that the following estimation

$$W(t) < \lambda_0 \epsilon_1 \epsilon_0^2 \quad (9)$$

holds for any $t \geq 0$.

For this purpose, we first show that (9) is valid for $t \in \Delta_{10}$. By the contrary, there exists at least one $t \in \Delta_{10}$ such that $W(t) \geq \lambda_0 \epsilon_1 \epsilon_0^2$. Let $t^* = \inf\{t \in [0, \delta) : W(t) \geq \lambda_0 \epsilon_1 \epsilon_0^2\}$. Observe that, $\forall \theta \in [-\bar{\tau}, 0]$,

$$W(\theta) = e^{\gamma \theta} \mu_1 x^T(\theta) P_{12} x(\theta) \leq \lambda_1 \mu_1 \|\phi\|_{\bar{\tau}}^2 < \lambda_0 \epsilon_1 \epsilon_0^2.$$

It follows that $t^* \in (0, \delta)$, $W(t^*) = \lambda_0 \epsilon_1 \epsilon_0^2$, $W(t) < \lambda_0 \epsilon_1 \epsilon_0^2$, $t \in [-\bar{\tau}, t^*)$, and

$$\dot{W}(t^*) \geq 0. \quad (10)$$

Note that $W(t^*) \geq W(t^* - \tau_{1t^*})$, and $\varphi(t) \in [\underline{\mu}_1, \bar{\mu}_1]$ for $t \in [-\bar{\tau}, \delta)$. For any $\alpha_1 > 0$, we have

$$\begin{aligned} 0 &\leq \alpha_1 \left(e^{\gamma \bar{\tau}} \bar{\mu}_1 x^T(t^*) \sum_{j=1}^2 \rho_j(t^*) P_{1j} x(t^*) \right. \\ &\quad \left. - \underline{\mu}_1 x^T(t^* - \tau_{1t^*}) \sum_{j=1}^2 \rho_j(t^* - \tau_{1t^*}) P_{1j} x(t^* - \tau_{1t^*}) \right) \\ &= \alpha_1 \sum_{j,h=1}^2 \rho_j(t^*) \tilde{\rho}_h(t^*) \left(e^{\gamma \bar{\tau}} \bar{\mu}_1 x^T(t^*) P_{1j} x(t^*) \right. \\ &\quad \left. - \underline{\mu}_1 x^T(t^* - \tau_{1t^*}) P_{1h} x(t^* - \tau_{1t^*}) \right). \end{aligned} \quad (11)$$

For $t \in (0, \delta)$, the differentiation of $W(t)$ along the trajectories of system (1) leads to

$$\begin{aligned} \dot{W}(t) &= e^{\gamma t} \varphi(t) \sum_{j=1}^2 \rho_j(t) \left\{ \left(\frac{\ln \mu_1}{\delta} + \gamma \right) x^T(t) P_{1j} x(t) \right. \\ &\quad \left. + \frac{1}{\delta} x^T(t) (P_{11} - P_{12}) x(t) + 2x^T(t) P_{1j} \mathcal{A}_1 \eta(t) \right\}. \end{aligned} \quad (12)$$

On the other hand, recalling assumption (A1), we obtain

$$\begin{aligned} 0 &\leq (\kappa_\ell^+ x_\ell - f_\ell(x_\ell)) (f_\ell(x_\ell) - \kappa_\ell^- x_\ell), \\ &\text{for any } x_\ell \in \mathbb{R}, \ell = 1, 2, \dots, n. \text{ It follows that, } \forall t \geq 0, \\ 0 &\leq \sum_{l=0}^1 \sum_{\ell=1}^n d_{\sigma(t)\ell}(t) (\kappa_\ell^+ x_\ell(t - \tau_{lt}) - f_\ell(x_\ell(t - \tau_{lt}))) \\ &\quad \times (f_\ell(x_\ell(t - \tau_{lt})) - \kappa_\ell^- x_\ell(t - \tau_{lt})) \\ &= \eta^T(t) \mathcal{E}_{0\sigma(t)}(t) \eta(t), \end{aligned} \quad (13)$$

where for $i = 1, 2$,

$$\mathcal{E}_{0i}(t) = \begin{bmatrix} L_1 D_{i0}(t) & 0 & L_2 D_{i0}(t) & 0 \\ * & L_1 D_{i1}(t) & 0 & L_2 D_{i1}(t) \\ * & * & -D_{i0}(t) & 0 \\ * & * & * & -D_{i1}(t) \end{bmatrix}.$$

Applying (11) and (13) to (12) with $t = t^*$ yields

$$\dot{W}(t^*) \leq e^{\gamma t^*} \varphi(t^*) \sum_{j,h=1}^2 \rho_j(t^*) \tilde{\rho}_h(t^*) \eta^T(t^*) \mathcal{E}_{1jh1} \eta(t^*).$$

Then, according to (7), we get $\dot{W}(t^*) < 0$. This contradicts (10). So (9) holds for $t \in \Delta_{10}$.

Suppose that the estimation (9) is satisfied for $t \in [-\bar{\tau}, t_{3-i,k+i-1})$ with any fixed $(i, k) \in \{1, 2\} \times \mathbb{N}$. Let us verify that it is also satisfied for $t \in \Delta_{3-i,k+i-1}$. Set $\tilde{t}_{i,k} = t_{3-i,k+i-1}$. According to the induction hypothesis, we have

$$\begin{aligned} \lambda_0 \epsilon_1 \epsilon_0^2 &> W(\tilde{t}_{i,k}^-) \\ &= e^{\gamma \tilde{t}_{i,k}} \varphi(\tilde{t}_{i,k}^-) x^T(\tilde{t}_{i,k}^-) \sum_{j=1}^2 \rho_j(\tilde{t}_{i,k}^-) P_{ij} x(\tilde{t}_{i,k}^-) \\ &= e^{\gamma \tilde{t}_{i,k}} \mu_i x^T(\tilde{t}_{i,k}^-) P_{i1} x(\tilde{t}_{i,k}^-). \end{aligned}$$

It follows from (6) that

$$\begin{aligned} W(\tilde{t}_{i,k}) &= e^{\gamma \tilde{t}_{i,k}} \varphi(\tilde{t}_{i,k}) x^T(\tilde{t}_{i,k}) \sum_{j=1}^2 \rho_j(\tilde{t}_{i,k}) P_{3-i,j} x(\tilde{t}_{i,k}) \\ &= e^{\gamma \tilde{t}_{i,k}} x^T(\tilde{t}_{i,k}) P_{3-i,2} x(\tilde{t}_{i,k}) \\ &\leq e^{\gamma \tilde{t}_{i,k}} \mu_i x^T(\tilde{t}_{i,k}) P_{i1} x(\tilde{t}_{i,k}) \\ &< \lambda_0 \epsilon_1 \epsilon_0^2. \end{aligned} \quad (14)$$

By the contrary, assume that (9) is not true for $t \in \Delta_{3-i,k+i-1}$. Then there exists a time $t \in (\tilde{t}_{i,k}, t_{i,k+1})$ such that $W(t) \geq \lambda_0 \epsilon_1 \epsilon_0^2$. Let $\bar{t} = \inf\{t \in \Delta_{3-i,k+i-1} : W(t) \geq \lambda_0 \epsilon_1 \epsilon_0^2\}$. In view of (14), we have that $\bar{t} \in (\tilde{t}_{i,k}, t_{i,k+1})$, $W(\bar{t}) = \lambda_0 \epsilon_1 \epsilon_0^2$, $W(t) < \lambda_0 \epsilon_1 \epsilon_0^2$, $t \in [-\bar{\tau}, \bar{t})$, and $\dot{W}(\bar{t}) \geq 0$. Assume that $\bar{t} - \tau_{1\bar{t}} \in \Delta_{p\bar{k}}$ for some $p \in \{1, 2\}$ and $\bar{k} \in \mathbb{N}$. Then we have $W(\bar{t}) > W(\bar{t} - \tau_{1\bar{t}})$, which implies that for any $\alpha_{3-i} > 0$,

$$\begin{aligned} 0 &\leq \alpha_{3-i} \left(e^{\gamma \bar{t}} \bar{\mu}_{3-i} x^T(\bar{t}) \sum_{j=1}^2 \rho_j(\bar{t}) P_{3-i,j} x(\bar{t}) \right. \\ &\quad \left. - \underline{\mu}_p x^T(\bar{t} - \tau_{1\bar{t}}) \sum_{j=1}^2 \rho_j(\bar{t} - \tau_{1\bar{t}}) P_{pj} x(\bar{t} - \tau_{1\bar{t}}) \right) \\ &= \alpha_{3-i} \sum_{j,h=1}^2 \rho_j(\bar{t}) \tilde{\rho}_h(\bar{t}) \left(e^{\gamma \bar{t}} \bar{\mu}_{3-i} x^T(\bar{t}) P_{3-i,j} x(\bar{t}) \right. \\ &\quad \left. - \underline{\mu}_p x^T(\bar{t} - \tau_{1\bar{t}}) P_{ph} x(\bar{t} - \tau_{1\bar{t}}) \right). \end{aligned}$$

Then applying (13) and the above inequality to $\dot{W}(\bar{t})$ yields

$$\dot{W}(\bar{t}) \leq e^{\gamma \bar{t}} \varphi(\bar{t}) \sum_{j,h=1}^2 \rho_j(\bar{t}) \tilde{\rho}_h(\bar{t}) \eta^T(\bar{t}) \mathcal{E}_{3-i,jhp} \eta(\bar{t}),$$

which leads to $\dot{W}(\bar{t}) < 0$ by using (7). This contradicts $\dot{W}(\bar{t}) \geq 0$. So (9) is true for $t \in \Delta_{3-i,k+i-1}$.

Based on the above analysis, we conclude that for any given $(i, k) \in \{1, 2\} \times \mathbb{N}$, the estimation (9) holds for $t \in [-\bar{\tau}, t_{3-i,k+i-1})$. It follows that (9) holds for all $t \geq -\bar{\tau}$. Thus, $\|x(t)\| \leq \sqrt{\epsilon_1/\mu_0} \|\phi\| e^{-\frac{\gamma}{2}t}$, $\forall t \geq 0$. This completes the proof. ■

Remark 2. Unlike the existing results, the condition of Theorem 1 is independent of both the size and the derivative of delay $\tau(t)$. This result is achieved by applying the piecewise switching-time-dependent Lyapunov function $V_1(t, x)$ given in (8). It can be seen from (14) that $V_1(t, x(t))$ is nonincreasing at switching instants. Moreover, inside the switching intervals, $\dot{V}_1(t, x(t)) < 0$ if certain Razumikhin-type condition is satisfied. These properties of $V_1(t, x(t))$ make the stability analysis act like that for the time-delay systems without switching, thereby leading to a delay-independent stability condition.

For Case 2, by making use of the information of the delay derivative, we will construct a piecewise switching-time-dependent Lyapunov functional for stability analysis of system (3).

Theorem 2. Given $m \times n$ matrix K , the control period ω , and the control width δ , consider system (3) satisfying assumptions (A1)–(A2). Under Case 2, if for some prescribed positive scalars μ_i , $i = 1, 2$, there exist $n \times n$ matrices $P_{ij} > 0$, $Q_j > 0$, $i, j = 1, 2$, $n \times n$ positive diagonal matrices D_{ijhl} , $i, j, h = 1, 2$, $l = 0, 1$, such that (6) and the

following LMIs hold:

$$\begin{bmatrix} \Omega_{2ijh} & P_{ij}C_1 & \Gamma_{ijh} & P_{ij}A_1 \\ * & \Phi_{ijh} & 0 & L_2D_{ijh1} \\ * & * & -D_{ijh0} & 0 \\ * & * & * & -D_{ijh1} \end{bmatrix} < 0, \quad i, j, h = 1, 2, \quad (15)$$

where Γ_{ijh} is defined in Theorem 1, and

$$\begin{aligned} \Omega_{2ijh} &= \frac{\ln \mu_i}{\delta_i} P_{ij} + \frac{1}{\delta_i} (P_{i1} - P_{i2}) + P_{ij}\bar{C}_{i0} + \bar{C}_{i0}^T P_{ij} \\ &\quad + \frac{1}{\underline{\mu}_i(1-r)} Q_j + L_1 D_{ijh0}, \\ \Phi_{ijh} &= -(1/\bar{\mu}_i) Q_h + L_1 D_{ijh1}, \end{aligned}$$

then system (3) is GES.

Proof. From (15), there exists a positive scalar γ such that LMIs (15) in which $\frac{\ln \mu_i}{\delta_i}$ is replaced by $\gamma + \frac{\ln \mu_i}{\delta_i}$ and $\frac{1}{\underline{\mu}_i(1-r)}$ is replaced by $\frac{e^{\gamma\bar{\tau}}}{\underline{\mu}_i(1-r)}$ still hold. For illustration purposes, the notation $\tilde{\Sigma}_{ijh}$ is used to denote the matrix on the left side of (15) in which $\frac{\ln \mu_i}{\delta_i}$ is replaced by $\gamma + \frac{\ln \mu_i}{\delta_i}$ and $\frac{1}{\underline{\mu}_i(1-r)}$ is replaced by $\frac{e^{\gamma\bar{\tau}}}{\underline{\mu}_i(1-r)}$. Then

$$\tilde{\Sigma}_{ijh} < 0, \quad i, j, h = 1, 2. \quad (16)$$

We introduce the following piecewise switching-time-dependent Lyapunov functional for system (3):

$$\begin{aligned} V(t, \phi) &= V_1(t, \phi(0)) \\ &\quad + \int_{t-\tau(t)}^t \frac{e^{-\gamma(t-s-\bar{\tau})}}{1-\dot{\tau}(\vartheta(s))} \phi^T(s-t) Q(s) \phi(s-t) ds, \end{aligned} \quad (17)$$

where $V_1(t, x)$ is defined in (8), $Q(t) = \sum_{j=1}^2 \rho_j(t) Q_j$, and $\vartheta(t)$ denotes the inverse of the strictly increasing function $t - \tau(t)$.

For any given initial function $\phi \in C([- \bar{\tau}, 0], \mathbb{R}^n)$, set $x(t) = x(t, 0, \phi)$, and $V(t) = V(t, x_t)$. For $t \in (t_{ik}, t_{3-i, k+i-1})$ with any fixed $(i, k) \in \{1, 2\} \times \mathbb{N}$, differentiating $V(t)$ along the trajectories of (3) and using the relation of $\underline{\mu}_i \leq \varphi(t) \leq \bar{\mu}_i$ leads to

$$\dot{V}(t) \leq -\gamma V(t) + \varphi(t) \sum_{j,h=1}^2 \rho_j(t) \tilde{\rho}_h(t) \eta^T(t) \tilde{\Sigma}_{ijh} \eta(t),$$

where inequality (13) is utilized. Then, according to (16), we obtain that $\dot{V}(t) \leq -\gamma V(t)$, for $t \in (t_{ik}, t_{3-i, k+i-1})$. It follows that for each $(i, k) \in \{1, 2\} \times \mathbb{N}$,

$$V(t) \leq V(t_{ik}) e^{-\gamma(t-t_{ik})}, \quad t \in \Delta_{ik}. \quad (18)$$

Next, we estimate $V(t)$ at switching instants t_{ik} , $(i, k) \in \{1, 2\} \times \mathbb{N}$. Note that condition (6) implies $V_1(t_{ik}) \leq V_1(t_{ik}^-)$, for all $(i, k) \in \{1, 2\} \times \mathbb{N}$. It follows that $V(t_{ik}) \leq V(t_{ik}^-)$, for all $(i, k) \in \{1, 2\} \times \mathbb{N}$. This fact, in combination with (18) yields $V(t) \leq V(0) e^{-\gamma t}$, for any $t \geq 0$, which concludes the proof. ■

Remark 3. The piecewise switching-time-dependent Lyapunov functional $V(t, \phi)$ given in (17) is quite different from those adopted in Chen et al. (2014) and Sun et al. (2008). One significant difference is that the switching-time-dependent weighting factor $\varphi(t)$ is introduced here to coordinate the dynamical behavior of the closed-loop and the open-loop modes. Under the condition of Theorem 2, the weighting factor $\varphi(t)$ can make the Lyapunov functional $V(t, x_t)$ along the trajectory of system (3) to have a negative derivative inside the switching intervals and to be nonincreasing at the switching instants. As a result, we obtain a new exponential stability condition which is independent of

the upper bound of the time delay. In contrast, the Lyapunov functional adopted in Chen et al. (2014) and Sun et al. (2008) decays during the closed-loop mode but diverges during the open-loop mode. This leads to the so-called “jump” phenomena of adjacent Lyapunov functionals at the switching instants. In order to eliminate the “jump” phenomena, the size of the time delay is not allowed to be very big, which results in the delay-dependent stability conditions therein.

Remark 4. From the viewpoint of intermittent control, the failure controller considered in Sun et al. (2008) is a delayed intermittent feedback controller. According to their intermittent control scheme, both the control period and the control width are not required to be fixed. Moreover, the designed intermittent controller can tolerate small input delays. Thus, the intermittent control scheme proposed in Sun et al. (2008) has the advantages of easier implementation and better robustness. Designing delayed intermittent controllers with non-fixed both control period and control width such that the intermittently controlled time-delay system is exponentially stable irrespective of the sizes of the state delay is a very meaningful topic which will be investigated in our future work.

Remark 5. Note that Case 2 can be treated as a special case of Case 1. This means that if the delays belong to Case 2, both Theorems 1 and 2 can be applied. However, as will be shown in Example 1, when r is sufficiently small, the stability criterion given in Theorem 2 is less conservative than that given in Theorem 1.

4. Controller design

In this section, under the assumption that

$$\kappa_i^- \leq 0 \leq \kappa_i^+, \quad i = 1, 2, \dots, n, \quad (19)$$

we will deal with the design problem of delay-independently periodically intermittent state-feedback controllers. For this purpose, we set $\bar{L}_1 = L_1^{\frac{1}{2}}$.

Theorem 3. Given the control period ω and the control width δ , consider the system (1) satisfying assumptions (A1)–(A2) and (19). Under Case 1, if for some prescribed positive scalars μ_i , α_i , ε_i , and θ_{ihp} , $i, h, p = 1, 2$, there exist $n \times n$ matrices X_0 , $X_{ij} > 0$, $i, j = 1, 2$, $n \times n$ diagonal matrices $\bar{D}_{ijhl} > 0$, $i, j, h = 1, 2, l = 0, 1$, and a $m \times n$ matrix $\bar{K}$, such that the following LMIs hold for all $h, p \in \{1, 2\}$:

$$\begin{bmatrix} \Lambda_{111hp} & \mathcal{L}_1^T(X_{11} - X_0^T + \theta_{1hp} B \bar{K}) \\ * & -\theta_{1hp}(X_0 + X_0^T) \end{bmatrix} < 0, \quad (20)$$

$$\begin{bmatrix} \Lambda_{112hp} & X_{12} \mathcal{L}_1^T & \mathcal{L}_1^T(X_{12} - X_0^T + \theta_{2hp} B \bar{K}) \\ * & -\delta_1 X_{11} & 0 \\ * & * & -\theta_{2hp}(X_0 + X_0^T) \end{bmatrix} < 0, \quad (21)$$

$$\Lambda_{121hp} < 0, \quad (22)$$

$$\begin{bmatrix} \Lambda_{122hp} & \mathcal{L}_1^T X_{22} \\ * & -\delta_2 X_{21} \end{bmatrix} < 0, \quad (23)$$

$$X_{21} \leq \mu_2 X_{12}, \quad X_{11} \leq \mu_1 X_{22}, \quad (24)$$

where $\mathcal{L}_1 = [I \ 0 \ 0 \ 0 \ 0 \ 0]$, and

$$\Lambda_{1ijhp} = \begin{bmatrix} \bar{\Omega}_{ij} + \alpha_i \bar{\mu}_i X_{ij} & C_1 X_{ph} & \bar{\Gamma}_{ijh} & \Upsilon_{1ijh} \\ * & -\alpha_i \bar{\mu}_i X_{ph} & 0 & \Upsilon_{2ijh} \\ * & * & -\bar{D}_{ijh0} & 0 \\ * & * & * & \bar{\Phi}_{1ijh} \end{bmatrix},$$

in which

$$\begin{aligned}\bar{F}_{ijh} &= A_0 \bar{D}_{ijh0} + X_{ij} L_2, \\ \gamma_{ijh} &= [A_1 \bar{D}_{ijh1} \quad X_{ij} \bar{L}_1 \quad 0], \\ \gamma_{2ph} &= [X_{ph} L_2 \quad 0 \quad X_{ph} \bar{L}_1], \\ \bar{\Phi}_{ijh} &= -\text{diag}\{\bar{D}_{ijh1}, \bar{D}_{ijh0}, \bar{D}_{ijh1}\},\end{aligned}$$

and

$$\begin{aligned}\bar{\Omega}_{11} &= \frac{\ln \mu_1 + 1 - 2\varepsilon_1}{\delta_1} X_{11} + \frac{\varepsilon_1^2}{\delta_1} X_{12} + C_0 X_{11} + B\bar{K} \\ &\quad + (C_0 X_{11} + B\bar{K})^T, \\ \bar{\Omega}_{12} &= \frac{\ln \mu_1 - 1}{\delta_1} X_{12} + C_0 X_{12} + B\bar{K} + (C_0 X_{12} + B\bar{K})^T, \\ \bar{\Omega}_{21} &= \frac{\ln \mu_2 + 1 - 2\varepsilon_2}{\delta_2} X_{21} + \frac{\varepsilon_2^2}{\delta_2} X_{22} + C_0 X_{21} + X_{21} C_0^T, \\ \bar{\Omega}_{22} &= \frac{\ln \mu_2 - 1}{\delta_2} X_{22} + C_0 X_{22} + X_{22} C_0^T,\end{aligned}$$

then the periodically intermittent control law (2) with $K = \bar{K} X_0^{-1}$ exponentially stabilizes system (1).

Proof. Define $P_{ij} = X_{ij}^{-1}$, $D_{ijhl} = \bar{D}_{ijhl}^{-1}$, $i, j, h = 1, 2, l = 0, 1$, and $K = \bar{K} X_0^{-1}$. It is easy to see that the matrix inequalities (24) are equivalent to (6). Using the Schur complement and the matrix inequalities: $-X_{11} X_{12}^{-1} X_{11} \leq -2\varepsilon_1 X_{11} + \varepsilon_1^2 X_{12}$, which is derived from Lemma 2, we obtain from (20)–(21) that

$$\begin{bmatrix} \tilde{\Psi}_{jhp} & \mathcal{D}^T(X_{1j} - X_0^T) + \varepsilon(\theta_{jhp} X_0) \\ * & -(\theta_{jhp} X_0) - (\theta_{jhp} X_0)^T \end{bmatrix} < 0, \quad j, h, p = 1, 2, \quad (25)$$

where $\tilde{\Psi}_{jhp} = \Psi_{jhp} + \varepsilon X_0 \mathcal{D} + \mathcal{D}^T X_0^T \varepsilon^T$, $\mathcal{D} = [I \quad 0 \quad 0 \quad 0]$, $\varepsilon = \text{col}\{BK, 0, 0, 0\}$, and

$$\begin{aligned}\Psi_{jhp} &= \begin{bmatrix} \Theta_{1jhp} & C_1 X_{ph} & A_0 \bar{D}_{1jh0} + X_{1j} L_2 & A_1 \bar{D}_{1jh1} \\ * & \Theta_{2jhp} & 0 & X_{ph} L_2 \\ * & * & -\bar{D}_{1jh0} & 0 \\ * & * & * & -\bar{D}_{1jh1} \end{bmatrix}, \\ \Theta_{1jhp} &= \left(\frac{\ln \mu_1}{\delta} + \alpha_1 \bar{\mu}_1 \right) X_{1j} + \frac{1}{\delta} X_{1j} (P_{11} - P_{12}) X_{1j} \\ &\quad + C_0 X_{1j} + X_{1j} C_0^T + X_{1j} L_1 D_{1j0} X_{1j}, \\ \Theta_{2jhp} &= -\alpha_1 \bar{\mu}_p X_{ph} + X_{ph} L_1 D_{1jh1} X_{ph}.\end{aligned}$$

In the light of Lemma 1, the matrix inequalities (25) imply that

$$\Psi_{jhp} + \varepsilon X_{1j} \mathcal{D} + \mathcal{D}^T X_{1j}^T \varepsilon^T < 0, \quad j, h, p = 1, 2. \quad (26)$$

Pre- and post-multiplying (26) by $\text{diag}\{P_{1j}, P_{ph}, D_{1jh0}, D_{1jh1}\}$ yields that the matrix inequalities (26) are equivalent to (5) with $i = 1$. Using the similar technique, it is easy to show that the matrix inequalities (22)–(23) imply (5) with $i = 2$. Hence, the conclusion of this theorem follows immediately from Theorem 1, which completes the proof. ■

Remark 6. The LMIs (5) contain the terms $P_{ij} B K$, $j = 1, 2$, which complicate the design of the gain matrix K . To solve for K , the implication relationship presented in Lemma 2 is used, which makes the condition in Theorem 3 more stringent than that in Theorem 1.

For Case 2, following a similar line as in the proof of Theorem 3, we can establish a delay-independently periodically intermittent stabilization criterion from Theorem 2, which is stated as follows.

Theorem 4. Given the control period ω and the control width δ , consider the system (1) satisfying assumptions (A1)–(A2) and (19). Under Case 2, if for some prescribed positive scalars μ_i , ε_i , and θ_{ih} , $i, h = 1, 2$, there exist $n \times n$ matrices $X_0, X_{ij} > 0$, $\bar{Q}_j > 0$, $i, j = 1, 2$, $n \times n$ positive diagonal matrices $\bar{D}_{ijhl}$, $i, j, h = 1, 2, l = 0, 1$, an $m \times n$ matrix $\bar{K}$, such that (24) and the following LMIs hold for all $h \in \{1, 2\}$:

$$\begin{bmatrix} \Lambda_{211h} & \mathcal{I}_2^T(X_{11} - X_0^T + \theta_{1h} B\bar{K}) \\ * & -\theta_{1h}(X_0 + X_0^T) \end{bmatrix} < 0, \quad (27)$$

$$\begin{bmatrix} \Lambda_{212h} & X_{12} \mathcal{I}_2^T & \mathcal{I}_2^T(X_{12} - X_0^T + \theta_{2h} B\bar{K}) \\ * & -\delta_1 X_{11} & 0 \\ * & * & -\theta_{2h}(X_0 + X_0^T) \end{bmatrix} < 0, \quad (28)$$

$$\Lambda_{221h} < 0, \quad (29)$$

$$\begin{bmatrix} \Lambda_{222h} & \mathcal{I}_2^T X_{22} \\ * & -\delta_2 X_{21} \end{bmatrix} < 0, \quad (30)$$

where $\mathcal{I}_2 = [\mathcal{I}_1 \quad 0]$, and

$$\Lambda_{2ijh} = \begin{bmatrix} \bar{\Omega}_{ij} & C_1 \bar{Q}_h & \bar{F}_{ijh} & \gamma_{2ijh} \\ * & -(1/\bar{\mu}_i) \bar{Q}_h & 0 & \gamma_{2h} \\ * & * & -\bar{D}_{ijh0} & 0 \\ * & * & * & \bar{\Phi}_{2ijh} \end{bmatrix},$$

in which $\bar{\Omega}_{ij}$, $i, j = 1, 2$, are defined in Theorem 3, and

$$\begin{aligned}\gamma_{2ijh} &= [A_1 \bar{D}_{ijh1} \quad X_{ij} \quad X_{ij} \bar{L}_1 \quad 0], \\ \gamma_{2h} &= [\bar{Q}_h L_2 \quad 0 \quad 0 \quad \bar{Q}_h \bar{L}_1], \\ \bar{\Phi}_{2ijh} &= -\text{diag}\{\bar{D}_{ijh1}, \bar{\mu}_i(1-r)\bar{Q}_i, \bar{D}_{ijh0}, \bar{D}_{ijh1}\},\end{aligned}$$

then the periodically intermittent control law (2) with $K = \bar{K} X_0^{-1}$ exponentially stabilizes system (1).

5. Numerical examples

In this section, two examples are provided to illustrate the effectiveness of the proposed results.

Example 1. Consider the periodically intermittently controlled time-delay system (3) with the following parameters:

$$\begin{aligned}C_0 &= \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}, \quad C_1 = 0, \quad A_0 = \begin{bmatrix} 1.8 & 1.5 \\ 0.1 & 1.8 \end{bmatrix}, \\ A_1 &= \begin{bmatrix} -1.4 & 0.1 \\ 0.1 & -1.4 \end{bmatrix}, \quad BK = \begin{bmatrix} -12 & 0 \\ 0 & -13 \end{bmatrix}.\end{aligned}$$

The nonlinear function is taken as $f(x) = \frac{1}{2}(|x+1| - |x-1|)$. Obviously, $\kappa_i^+ = 1$, $\kappa_i^- = 0$, $i = 1, 2$. Thus, $L_1 = 0$, and $L_2 = 0.5I_2$. For different control periods, we apply Theorems 1 and 2 to calculate the minimum control widths, which preserve the stability irrespective of the size of $\tau(t)$. The obtained results are listed in Table 1.

The calculation results in Table 1 show that when $r \leq \frac{1}{2}$, Theorem 2 can provide smaller control width than Theorem 1. However, with the values of r increasing, Theorem 2 becomes more conservative than Theorem 1. On the other hand, the calculation results also reveal that the derived largest open-loop widths, i.e.,

Table 1The minimum control width $\delta_{\min}$ for $\dot{\tau}(t) \leq r$.

ω	10	5	3	2	1
Theorem 1 (unknown r)	9.79	4.79	2.79	1.79	0.79
Theorem 2 ($r = 0$)	9.69	4.69	2.70	1.70	0.70
Theorem 2 ($r = \frac{1}{4}$)	9.73	4.73	2.73	1.73	0.74
Theorem 2 ($r = \frac{1}{2}$)	9.78	4.78	2.78	1.78	0.78
Theorem 2 ($r = \frac{2}{3}$)	9.82	4.82	2.82	1.82	0.82

Table 2The upper bound $\bar{\tau}_{\max}$ for $r = 0$.

ω	10	5	3	2	1
Hu et al. (2010)	4.94	2.42	1.41	0.9	0.38
Yu et al. (2011)	4.68	2.28	1.32	0.84	0.34
Xing et al. (2013)	2.67	1.29	0.74	0.47	0.19
Chen et al. (2014)	6.86	3.43	2.03	1.3	0.55
Theorem 2	Any	Any	Any	Any	Any

Table 3The upper bound $\bar{\tau}_{\max}$ for unknown r .

ω	10	5	3	2	1
Hu et al. (2010)	N/A	N/A	N/A	N/A	N/A
Yu et al. (2011)	4.68	2.28	1.32	0.84	0.34
Xing et al. (2013)	3.57	1.74	1.01	0.64	0.27
Chen et al. (2014)	2.34	1.11	0.66	0.4	0.09
Theorem 1	Any	Any	Any	Any	Any

$\omega - \delta_{\min}$, for different ω , are pretty much the same. For example, when r is chosen to be 0, $\frac{1}{4}$, $\frac{1}{2}$, and “unknown”, successively, the corresponding upper bound of $\omega - \delta$ is about 0.30, 0.27, 0.22, and 0.21, respectively.

The stability conditions in Chen et al. (2014), Hu et al. (2010), Xing et al. (2013) and Yu et al. (2011) depend on the delay bound $\bar{\tau}$. For the case of $r = 0$ and the case with unknown r , when (ω, δ) is restricted to be the values given in Table 1, the maximum allowable upper bound $\bar{\tau}_{\max}$ achieved by those methods as well as our Theorem 1 and Theorem 2 are summarized in Table 2 and Table 3, respectively. The calculation results show that when the control periods are small, e.g., $\omega \leq 2$, those related results are only suitable to the time-delay systems with small state delays. On the other hand, when δ/ω is close to 1, our results are less conservative than the existing results.

With the initial condition of $x(\theta) = (1, -1)^T$, the response curves of the intermittently controlled system with the control period $\omega = 1$ and the control width $\delta = 0.7$ are depicted in Fig. 1(a) and (b) for the time-delay $\tau(t) = 2$ and $\tau(t) = 10$, respectively. The simulation results show that they fit well with the theoretical predications.

Example 2. Consider a controlled time-delay system (1) with the following parameters:

$$C_0 = \begin{bmatrix} -am_1 & a & 0 \\ 1 & -1 & 1 \\ 0 & -b & 0 \end{bmatrix}, \quad C_1 = \begin{bmatrix} -c & 0 & 0 \\ -c & 0 & 0 \\ 2c & 0 & -c \end{bmatrix},$$

$$A_0 = \begin{bmatrix} a(m_1 - m_0) & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad A_1 = 0, \quad B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix},$$

where $a = 9$, $b = 14.286$, $c = 0.1$, $m_0 = -\frac{1}{7}$, $m_1 = \frac{2}{7}$, and the nonlinear function is taken as $f(x) = (f_1(x_1), f_2(x_2), f_3(x_3))^T$ with $f_i(x_i) = \frac{1}{2}(|x_i + 1| - |x_i - 1|)$, $i = 1, 2, 3$. When $u = 0$,

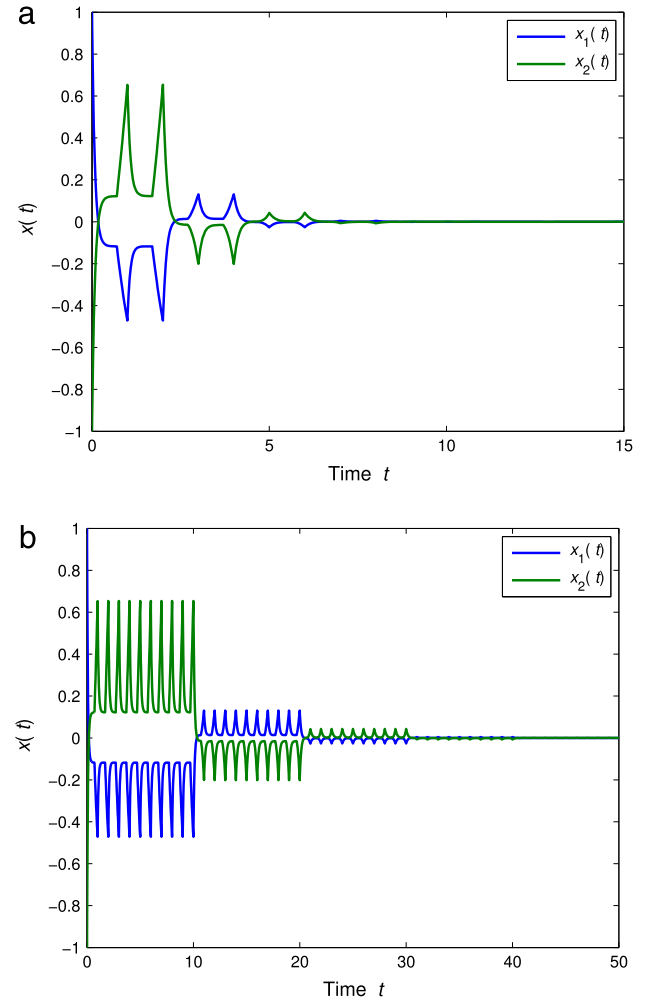


Fig. 1. Simulations of state responses for $\omega = 1$ and $\delta = 0.7$: (a) $\tau(t) = 2$; (b) $\tau(t) = 10$.

this time-delay system represents the model of time-delay Chua's circuit (He, Qian, Cao, & Han, 2011). In the case that $\tau(t) = 1$, with the initial condition $\phi(s) = (0.2, 0.1, -0.5)^T$ for all $s \in [-1, 0]$, the time-delay Chua's circuit exhibits chaotic behavior, as shown in Fig. 2. In the following, we distinguish two cases to design delay-independently intermittent controllers.

Case 1: $\omega = 1$ and $\tau(t)$ is a time-varying bounded function.

By applying Theorem 3 and solving the LMIs (20)–(24), with the choice of $\mu_1 = 2.89$, $\mu_2 = 0.24$, $\varepsilon_1 = 0.92$, $\varepsilon_2 = 1.31$, $\alpha_1 = 0.32$, $\alpha_2 = 3.44$, $\theta_{1hp} = 0.008$, and $\theta_{2hp} = 0.026$, $h, p = 1, 2$, the smallest obtainable δ is $\delta_{\min} = 0.83$, and the corresponding controller gain matrix is given by $K = [-58.2510 \quad -156.3118 \quad -9.1416]$.

Case 2: $\omega = 1$ and $\tau(t) = \text{const}$.

Applying Theorem 4 and choosing $\mu_1 = 2.62$, $\mu_2 = 0.31$, $\varepsilon_1 = 0.9$, $\varepsilon_2 = 1.38$, $\theta_{1h} = 0.008$, and $\theta_{2h} = 0.026$, $h = 1, 2$, it has been found that the minimum value of δ such that the LMIs (27)–(30) and (24) are feasible is $\delta_{\min} = 0.69$. The corresponding gain matrix is given by $K = [-134.7355 \quad -210.2626 \quad -44.7893]$. In the simulation, the initial state is set to $x(\theta) = (3, -1, -1.5)^T$, $\theta \in [-\bar{\tau}, 0]$. The time responses for $\tau(t) = 2$ and for $\tau = 10$ are plotted in Fig. 3(a) and (b), respectively. It is seen that the same periodically intermittent controller indeed asymptotically stabilizes the zero solution for different delays.

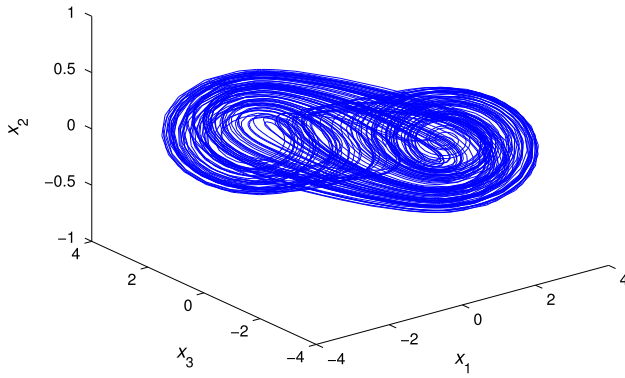


Fig. 2. Chaotic attractor of time-delay Chua's circuit.

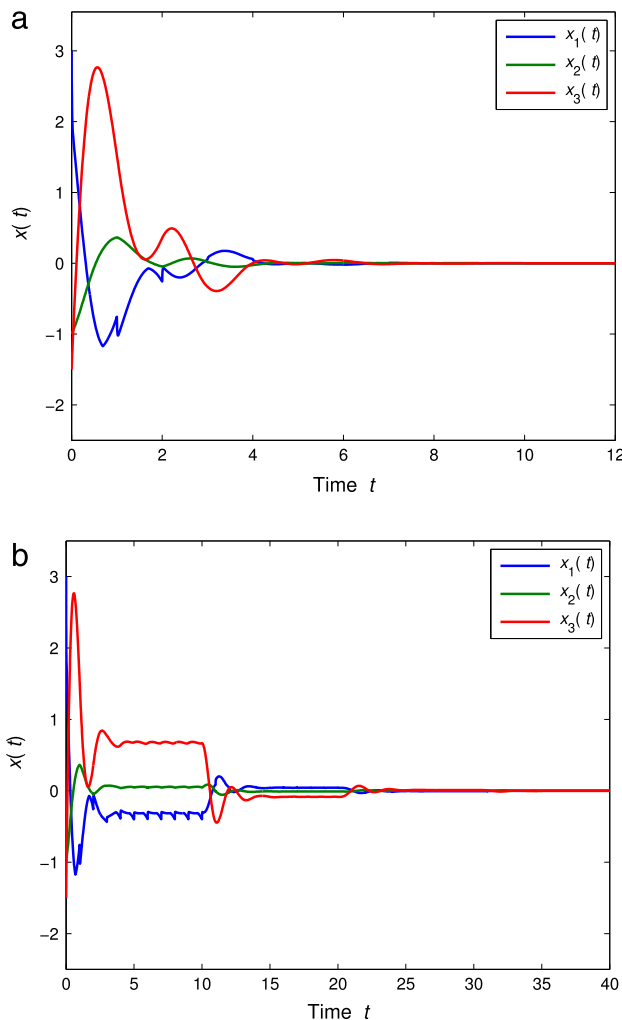


Fig. 3. Simulations of state responses under periodically intermittent control: (a) $\tau(t) = 2$; (b) $\tau(t) = 10$.

6. Conclusion

In this paper, we have addressed the problem of delay-independently periodically intermittent stabilization for a class of time-delay systems. This problem has been resolved by introducing the novel piecewise switching-time-dependent Lyapunov functions/functionals. Such Lyapunov functions/functionals can efficiently eliminate the effects of the size of time-delay on the system stability and enable the establishment of delay-independent

exponential stability criteria for periodically intermittent controlled time-delay systems. Based on these stability criteria, the new methods for designing periodically intermittent state feedback controllers have been presented, which possess the same merits as the stability criteria. The capability of the proposed methods has been confirmed by the two numerical examples.

In the intermittent control framework, the control input is zeroed out during the period that the controller does not work. This control scheme may degrade system performance. A possible method for overcoming such weakness is to introduce a zero-order-holder in the controlled system such that the actuator can hold its current value when the control signal is not available. The resulting closed-loop system should be a switched system consisting of a continuously controlled subsystem and a sampled-data subsystem. An interesting future work may be to apply the switching-time-dependent Lyapunov function/functional based technique to establish the stabilization criterion.

References

- Chen, W.-H., & Zheng, W.X. (2007). Robust stabilization of delayed Markovian jump systems subject to parametric uncertainties. In *Proceedings of the 46th IEEE conference on decision and control, New Orleans, LA, December 2007* (pp. 3054–3059).
- Chen, W.-H., & Zheng, W. X. (2010). Delay-independent minimum dwell time for exponential stability of uncertain switched delay systems. *IEEE Transactions on Automatic Control*, 55(10), 2406–2413.
- Chen, W.-H., & Zheng, W. X. (2012). An improved stabilization method for sampled-data control systems with control packet loss. *IEEE Transactions on Automatic Control*, 57(9), 2378–2384.
- Chen, W.-H., Zhong, J., Jiang, Z., & Lu, X. (2014). Periodically intermittent stabilization of delayed neural networks based on piecewise Lyapunov functions/functionals. *Circuits, Systems and Signal Processing*, 33(12), 3757–3782.
- Estrada, T., & Antsaklis, P. J. (2010). Model-based control with intermittent feedback: bridging the gap between continuous and instantaneous feedback. *International Journal of Control*, 83(12), 2588–2605.
- Garcia, E., & Antsaklis, P.J. (2013). Model-based Control of continuous-time systems with limited intermittent feedback. In *Proceedings of the 21st Mediterranean conference on control and automation, MED, Plataniass-Chania, Crete, Greece, June 2013* (pp. 452–457).
- Gawthrop, P. J., Neild, S. A., & Wagg, D.J. (2012). Semi-active damping using a hybrid control approach. *Journal of Intelligent Material Systems and Structures*, 23(18), 2103–2116.
- Gawthrop, P. J., & Wang, L. (2009a). Constrained intermittent model predictive control. *International Journal of Control*, 82(6), 1138–1147.
- Gawthrop, P. J., & Wang, L. (2009b). Event-driven intermittent control. *International Journal of Control*, 82(12), 2235–2248.
- He, W., Qian, F., Cao, J., & Han, Q.-L. (2011). Impulsive synchronization of two nonidentical chaotic systems with time-varying delay. *Physics Letters A*, 375(3), 498–504.
- Hu, C., Yu, J., Jiang, H., & Teng, Z. (2010). Exponential stabilization and synchronization of neural networks with time-varying delays via periodically intermittent control. *Nonlinearity*, 23(10), 2369–2390.
- Huang, J., Li, C., & Han, Q. (2009). Stabilization of delayed chaotic neural networks by periodically intermittent control. *Circuits, Systems and Signal Processing*, 28(4), 567–579.
- Huang, T., Li, C., & Liu, X. (2008). Synchronization of chaotic systems with delay using intermittent linear state feedback. *Chaos*, 18(3), 033122.
- Huang, T., Li, C., Yu, W., & Chen, G. (2009). Synchronization of delayed chaotic systems with parameter mismatches by using intermittent linear state feedback. *Nonlinearity*, 22(3), 569–584.
- Li, C., Liao, X., & Huang, T. (2007). Exponential stabilization of chaotic systems with delay by periodically intermittent control. *Chaos*, 17(1), 013103.
- Liu, X., Shen, X., & Zhang, H. (2011). Intermittent impulsive synchronization of chaotic delayed neural networks. *Differential Equations and Dynamical Systems*, 19(1–2), 149–169.
- Ronco, E., Arsan, T., & Gawthrop, P. J. (1999). Open-loop intermittent feedback control: Practical continuous-time GPC. *IEE Proceedings-Control Theory and Applications*, 146(5), 426–434.
- Song, Q., & Huang, T. (2015). Stabilization and synchronization of chaotic systems with mixed time-varying delays via intermittent control with non-fixed both control period and control width. *Neurocomputing*, 154, 61–69.
- Sun, X.-M., Liu, G.-P., Rees, D., & Wang, W. (2008). Stability of systems with controller failure and time-varying delay. *IEEE Transactions on Automatic Control*, 53(10), 2391–2396.
- Sun, X.-M., Zhao, J., & Hill, D. J. (2006). Stability and L_2 -gain analysis for switched delay systems: A delay-dependent method. *Automatica*, 42(10), 1769–1774.
- Wang, Y., Hao, J., & Zuo, Z. (2010). A new method for exponential synchronization of chaotic delayed systems via intermittent control. *Physics Letters A*, 374(19–20), 2024–2029.

- Wen, G., Duan, Z., Ren, W., & Chen, G. (2014). Distributed consensus of multi-agent systems with general linear node dynamics and intermittent communications. *International Journal of Robust and Nonlinear Control*, 24(16), 2438–2457.
- Xing, J., Jiang, H., & Hu, C. (2013). Exponential synchronization for delayed recurrent neural networks via periodically intermittent control. *Neurocomputing*, 113, 122–129.
- Yang, X., & Cao, J. (2009). Stochastic synchronization of coupled neural networks with intermittent control. *Physics Letters A*, 373(36), 3259–3272.
- Yu, J., Hu, C., Jiang, H., & Teng, Z. (2011). Exponential synchronization of Cohen–Grossberg neural networks via periodically intermittent control. *Neurocomputing*, 74(10), 1776–1782.
- Zhang, W.-A., & Yu, L. (2010). Stabilization of sampled-data control systems with control inputs missing. *IEEE Transactions on Automatic Control*, 55(2), 447–452.
- Zhao, J., & Hill, D. J. (2008). On stability, L_2 -gain and H_∞ control for switched systems. *Automatica*, 44(5), 1220–1232.
- Zochowski, M. (2000). Intermittent dynamical control. *Physica D*, 145(3–4), 181–190.



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